

PROJECT REPORT

**Data Fundamentals with Data Warehousing, SQL
and
Python**

B.Tech Program in Computer Science Engineering

Submitted By

Name: Sakshi Sahoo

Roll No: 23052348

**Kalinga Institute of Industrial Technology
(Deemed to be University)
Bhubaneswar, India**

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Title : Employee Salary Data Analysis

I) Problem Statement :

Organizations maintain large volumes of employee data including department, experience, salary, and location. However, without proper analysis, this data cannot provide useful insights.

This project aims to analyze employee salary data to understand salary distribution across departments and evaluate how experience influences salary levels, helping management in better decision-making.

II) Proposed Solutions / Features :

The proposed solution follows a multi-tool approach for analysis and visualization:

- SQL is used for structured data querying
- Excel is used for summarization using Pivot Tables
- Python is used for data analysis and visualization

The system extracts key insights and presents them in a simple and clear format.

Key Features

1. Department-wise Salary Analysis

The system identifies average salary across departments such as Finance, HR, and IT using SQL queries and Python analysis. This helps understand which departments are better compensated.

2. Experience vs Salary Relationship

By analyzing experience levels against salary figures, the project shows a positive correlation — employees with more experience tend to earn higher salaries.

3. Highest Paid Employee Identification

The project identifies the highest-paid employee using SQL queries, enabling management to benchmark salary levels.

4. Average Salary Calculation

Statistical measures such as average salary are computed to understand the central tendency of salary distribution across the organization.

5. Excel-Based Summary (Pivot Table)

An Excel pivot table is used to summarize department-wise average salaries, providing a quick and interactive way to explore data.

6. Multi-Tool Integration

The project integrates:

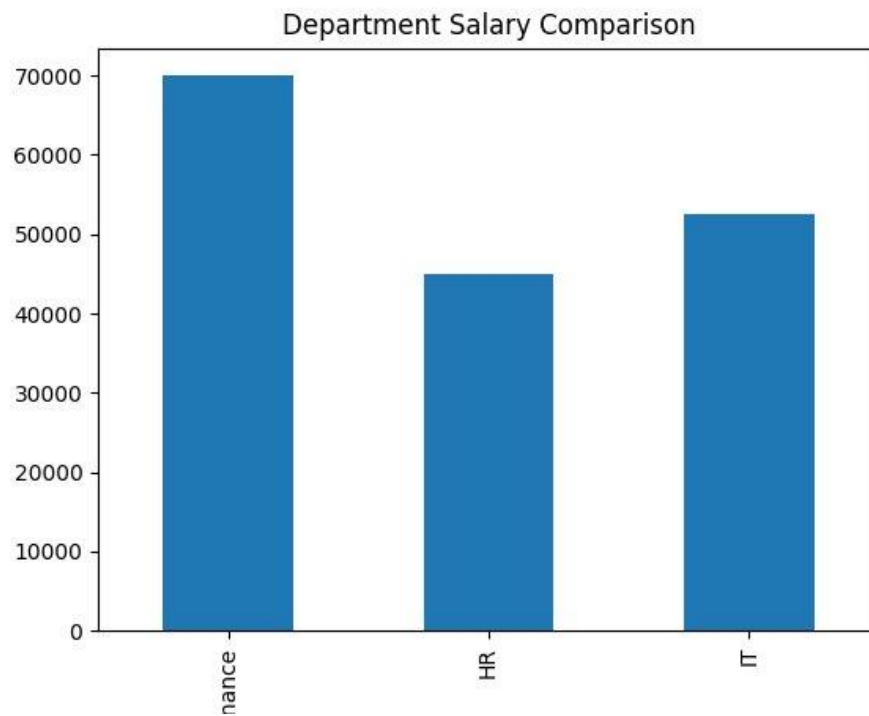
- SQL for querying
- Excel for summarization
- Python for analysis and visualization

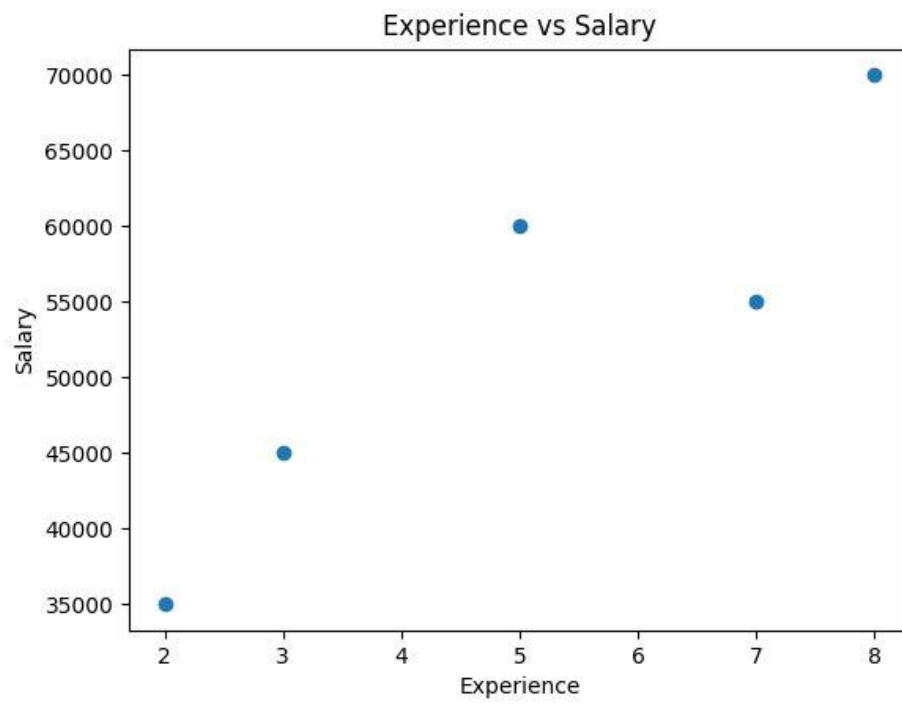
III) Screenshots:

- **SQL Outputs :**

	emp_id	name	department	experience	salary	city
0	101	Anil	IT	5	60000	Pune
1	102	Sunita	HR	7	55000	Mumbai
2	103	Rohit	IT	3	45000	Pune
3	104	Neha	Finance	8	70000	Nagpur
4	105	Ajay	HR	2	35000	Mumbai

- **Python Outputs :**





IV) Tech Stack :

The project utilizes a combination of tools and technologies to perform data analysis, querying, and visualization effectively. Each component plays a specific role in extracting and presenting insights from the employee dataset.

1. Python

Python is used as the primary programming language for data analysis and visualization. It provides powerful libraries that simplify data processing and graphical representation.

- Pandas: Used for data loading, cleaning, and analysis
- Matplotlib: Used for creating visualizations such as bar charts and scatter plots

2. SQL (MySQL)

SQL is used for structured data querying and analysis. The project uses MySQL Workbench to execute queries and extract insights from the dataset.

- Data storage using relational tables
- Aggregation using GROUP BY
- Statistical queries like AVG() and MAX()
- Identifying trends such as highest paid employee and average salary

3. Microsoft Excel

Microsoft Excel is used for quick data summarization and interactive analysis.

- Creation of Pivot Tables
- Department-wise average salary analysis
- Easy visualization of summarized data

4. CSV (Comma-Separated Values)

The dataset is stored in CSV format, which serves as a lightweight and portable data source.

- Easy to import into Python, SQL, and Excel
- Structured tabular format for consistency

V) Unique Points :

This project stands out due to its practical approach and integration of multiple tools to perform end-to-end data analysis. Unlike basic projects that rely on a single technology, this solution combines SQL, Excel, and Python to demonstrate a complete data analytics workflow.

Key Unique Aspects

1. Multi-Tool Integration

The project effectively integrates SQL for querying, Excel for summarization, and Python for analysis and visualization, showcasing versatility in handling data across platforms.

2. Real-World Use Case

The dataset and analysis are based on a real-world HR scenario, making the project relevant and applicable to organizational management systems.

3. Insight-Driven Approach

Instead of just displaying data, the project focuses on extracting meaningful insights such as salary distribution by department, experience-salary correlation, and highest-paid employee identification.

4. Visual Representation of Data

The use of charts (bar chart and scatter plot) makes complex data easy to understand, enabling better decision-making.

5. Structured and Scalable Design

The project is organized in a modular way (CSV → SQL → Excel → Python), making it easy to scale or extend for larger datasets and real-time systems.

VI) Future Improvements :

1. Machine Learning Integration — Implement predictive models to forecast salaries based on employee history and trends, enabling proactive HR decisions.
2. Interactive Dashboard — Develop a web-based dashboard using tools like Streamlit or Power BI for real-time data visualization and user interaction.
3. Real-Time Data Integration — Connect the system to live HR databases to analyze data dynamically instead of using static CSV files.
4. Employee Segmentation Analysis — Extend the project to analyze employees based on age groups, gender, or experience brackets for more detailed insights.
5. Automated Reporting System — Generate automated reports for management, reducing manual effort and improving efficiency.

VII) Key Insights :

- Finance department has the highest average salary (~₹70,000)
- HR department has comparatively lower salaries (~₹45,000)

- Average salary across all employees is ₹53,000
- Highest-paid employee is Neha with a salary of ₹70,000
- Employees with more experience tend to earn higher salaries
- Positive correlation exists between experience and salary

VIII) Conclusion :

This project demonstrates how employee data can be analyzed to generate meaningful insights. By combining SQL, Excel, and Python, it provides a complete data analysis workflow that can help organizations understand salary trends and improve decision-making. The visualizations and SQL outputs clearly show the salary distribution across departments and the positive relationship between experience and salary, making this a useful tool for HR analytics.